

$K[x]$ R unit. comm. ring

$I \subseteq R$ is called an ideal in R if:

① $0 \in I$

② $a, b \in I \Rightarrow a+b \in I$

③ $a \in I, z \in R \Rightarrow a \cdot z \in I$

$\forall f(x) \in K[x]$

$$f(x) = a_0 x^m + a_1 x^{m-1} + \dots + a_n$$

$$a_i \in K$$

Examples of fields:

TO BE FOUND

Objective:
Understand ideals
in $R = K[x]$
(where K is a field)

Long division
set up

if $f, g \in K[x], g \neq 0$
then there exists always q, r
such that $f = qg + r$ and
unique and following equation
holds. $\deg r < \deg g$

$$f = qg + r$$

$\deg 0 = -\infty$ makes sense

makes
sense

RUSH

How do you give someone a
concrete ideal in $K[x]$?

Examples:

$R = \mathbb{Z}$

$R = K[x]$

$$I = 3\mathbb{Z} = \mathbb{Z}$$

①, ②, ③ satisfied.

$K = \text{TO BE FIXED}$
 TO BE FOUND

Hilbert's basis theorem:

If R is Noetherian ring then $R[x]$ is also Noetherian.

$\leadsto R[x_1, \dots, x_n]$ also Noetherian because $R[x_1, \dots, x_{n-1}][x_n]$

finitely generated such that $I_1 \subset I_2 \subset I_3 \dots \underbrace{I_r = I_{r+1}}_{\text{stabilisation}}$

Ryza read
Serge Lang
!!!
...

$$R \Rightarrow R[x]$$

$$R \Leftarrow R[x] \Leftarrow$$

Ideals in R
are finitely
generated

R is Noetherian

Is $\mathbb{Z}$ Noetherian?

$6\mathbb{Z} \subset 3\mathbb{Z} \subset \mathbb{Z} \subset \dots$
 $I_1 \subset I_2 \subset I_r$

$\langle f_1 \rangle \subsetneq I \subsetneq R$
 $\uparrow$
 $\langle f_1, f_2 \rangle$
ideal noetherian

Stronger version than
in the lecture.

Def. A unitary comm. ring R
is called Noetherian if
for every ideals

$$I_1 \subset I_2 \subset \dots \subset I_k \subset \dots$$

in this ring there is some
 $r \in \mathbb{N}$ such that

$$I_r = I_{r+1} = I_{r+2} = \dots$$

(that is every ascending chain of
ideals stabilizes).

$$R = \mathbb{Z}$$

$$R[x] \subsetneq R[x] \subsetneq (x^2)$$

$$I_1 \subsetneq 2x$$

$$I_2 \subsetneq x^2$$

$$\mathbb{Z}[x]$$

$$\langle 2x, x^2 \rangle \subsetneq \mathbb{Z}[x]$$

$$I_1 \subsetneq I_2 \subsetneq I_r$$

CRT $\Rightarrow$ RSA

CRT:

Let I and J be coprime ideals in a Ring R . $I + J = R$.

$$R/I \cap J \rightarrow R/I \times R/J$$

$$\psi([a]_{I \cap J}) = ([a]_I, [a]_J), a \in R$$

$$R = \mathbb{Z}, I = p\mathbb{Z}, J = q\mathbb{Z}$$

$$\textcircled{R/I} \quad x^p \equiv x \pmod{p} \Leftrightarrow x^{(p-1)} \equiv 1 \pmod{p} \Leftrightarrow x^{(p-1)(q-1)} \equiv 1 \pmod{p}$$

$$\textcircled{R/J} \quad y^q \equiv y \pmod{q} \Leftrightarrow y^{(q-1)} \equiv 1 \pmod{q} \Leftrightarrow y^{(p-1)(q-1)} \equiv 1 \pmod{q}$$

$$\Leftrightarrow \begin{cases} x^{e \cdot m(p-1)(q-1)} \equiv x^p \pmod{p} \\ y^{e \cdot m(p-1)(q-1)} \equiv y^q \pmod{q} \end{cases} \xrightarrow{\text{CRT}} \begin{cases} x^e \equiv x \pmod{p} \\ y^e \equiv y \pmod{q} \end{cases}$$

Haring $x=0$?
Haring $y=0$?

(x, y)

RSA: Math problem.
 p, q , prime number.

$$n = p \cdot q$$

Key generation:

- take pair of prime numbers, $p \neq q$
- compute $n = p \cdot q$
- compute $\phi(n) = (p-1)(q-1)$
- we can find e , $\gcd(e, \phi(n)) = 1$
- we can find $d \equiv e^{-1} \pmod{\phi(n)}$
- $\text{pub} = (e, n)$, $\text{priv} = (d, n)$

Encryption:

- we can choose $m < n$, with $\text{priv} = (e, n)$
- compute $c \equiv m^e \pmod{n}$

Decryption:

- we have c , $\text{priv} = (d, n)$
- $m \equiv c^d \pmod{n}$

$$x^{(p-1)(q-1)} \equiv 1 \pmod{p} \Leftrightarrow x^{m(p-1)(q-1)} \equiv 1 \pmod{p}$$

$$\text{pow}_e(m) = \text{pow}_{e+m(p-1)(q-1)}(m)$$

$$\text{int}(\mathbb{Z}/pq\mathbb{Z}) = R/I \cap J$$

Hajji